

Effects of Scene Visual Characteristics on the Perception of Rotation Gains

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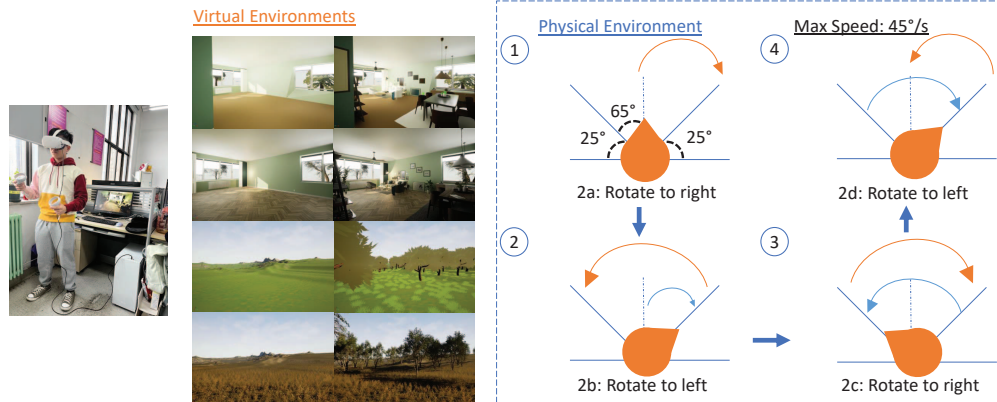


Figure 1: Eight VR experimental conditions consisted of two levels of visual density (high, low), spatial sizes (landscape, room), and realism (high, low).

ABSTRACT

Rotation gain is a crucial technique in Redirected Walking (RDW) for altering users' virtual trajectories. However, prior research [3] lacked a systematic exploration of how different visual characteristics impact users' perception of rotation speed. Therefore, this study proposes an investigation into this issue by designing eight VR scenes varying in visual density, spatial size, and realism. Through a within-subject study, the impact of these factors on the subjective detection of rotational speed was assessed. The results revealed significant differences in perceptual detection thresholds for rotation gains between room and landscape scenes, particularly in the lower bound of rotation gain thresholds.

Index Terms: Computing methodologies—Computer graphics—Graphics system and interface—Virtual Reality; Computing methodologies—Computer graphics—Graphics system and interface—Perception

1 INTRODUCTION

Rotation gains [4] have consistently been the most commonly employed subtle manipulation technique due to their superior capability in manipulating a user's virtual trajectory compared to translation gains. Previous literature has shown that VR scene visual characteristics can influence the user's translation perception, affecting metrics such as the Low Detection Threshold (LDT), Point of Subjective Equality (PSE), and High Detection Threshold (HDT) [2]. While some studies have concluded that rotation gain thresholds do not vary significantly with different visual densities [3], further investigation is needed into other aspects of the scene's visual characteristics. Therefore, this work designed eight virtual scenes varying in three aspects to study user subjective detection in rotation gains.

2 METHODS AND MATERIALS

We recruited a sample size of 29 participants, comprising 13 females and 16 males, with diverse backgrounds, including varying ages,

occupations, and education levels. All participants successfully completed the study, and none had any severe physical or mental conditions that would make them unsuitable for the experiment.

2.1 Conditions and Task

Our experimental conditions (blocks) encompassed eight distinct VR scenes (depicted in Figure 1 middle). Visual density is parameterized by the number of objects [3]. All objects are placed 50 units away from the VR avatar in all the high-visual density scenes. The configuration of spatial sizes was guided by Kim et al.'s [2] definition of angle of decline (AoD). The two levels of scene realism differ in terms of texture details, geometry fidelity, and shadows.

In each experimental block, we recorded the user's age, gender, and level of prior experience with VR using a questionnaire. All users were required to complete an SSQ at the beginning of the main experiment. We then explained the experimental procedure and objectives to them. The entire experiment commenced after we assisted each user in adjusting the interpupillary distance (IPD) and donning the HMD. The tested gains were at only seven levels (ranging from 0.7 to 1.3). Prior to this, each participant completed five practice trials (with gains of 1.0, 0.7, 1.0, 1.3, and 1.0) to familiarize themselves with the stimulus; correct answers were provided to the participants for reference after these trials. Subsequently, they took part in the formal experiments and provided their subjective responses for the 42 trials (6 repetitions $\times$ 7 levels of gains) for each scene.

The task of each trial is illustrated in Figure 1 right. Users were limited to rotating four times with a magnitude of $+65^\circ$ (right), -130° (left), $+130^\circ$ (right), and -65° (left). Throughout the rotation, the maximum rotation speed limit was set at $45^\circ/\text{s}$. After the gain-experience task, participants would answer the 2AFC question, whether they felt their yaw axis rotational speed in the virtual environment was "greater" or "slower" compared to the real environment. We utilized an Oculus Quest 2 headset with controllers as our primary experimental equipment. The eight VR scenes were developed in Unreal Engine 5.1 64-bit, integrated with the Unreal OpenXR Interface. Both the left-hand and right-hand controllers were utilized for experimental interactions. Participants used the 'A', 'B', 'X', and 'Y' buttons to respond to 2AFC's questions.

After each block, the users were asked to complete the Kennedy-Lane Simulator Sickness Questionnaire (SSQ) [1]. Each participant was required to complete two pseudo-random blocks (scenes) per day

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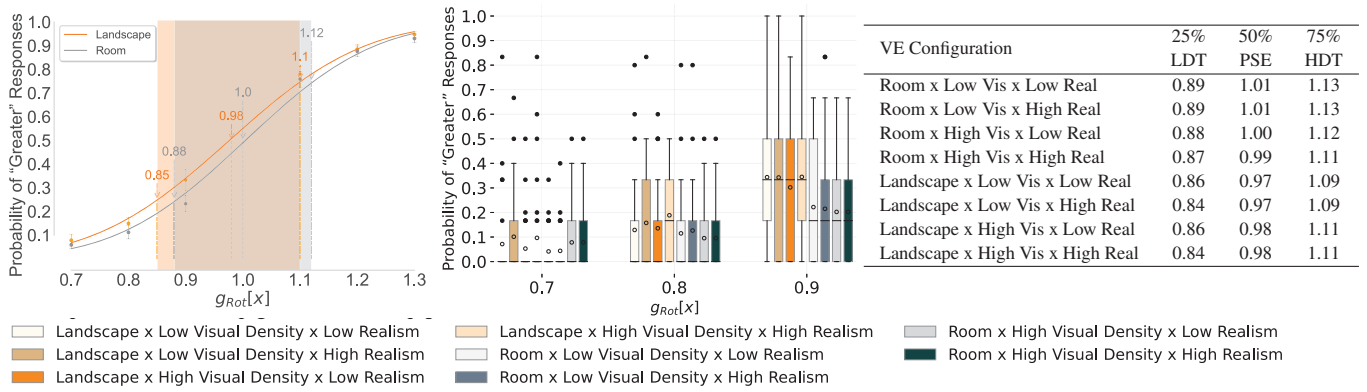


Figure 2: The effects of scene spatial size on the psychometric curve and rotation gain thresholds. 'Vis' denotes visual density; 'Real' denotes Realism.

over four consecutive days to mitigate the interference of simulation sickness. Compensation of 20 per hour was provided.

2.2 Data Analysis

We performed data analysis with a total of 9744 trials (29 users $\times$ 8 scenes $\times$ 7 gains $\times$ 6 repeats) in two aspects: participants' subjective detection and psychometric curve analysis using [5]'s method. For each participant, we first computed the probability of choosing 'greater' at each level of gain. Then, we conducted a 5-way ANOVA multivariate analysis (visual density + spatial size + realism + gain value + gender) to test for significant differences. As the Shapiro-Wilk test for normality indicated that the data were normally distributed, post-hoc analyses were conducted using pairwise t-tests with Bonferroni corrections. Lastly, the fitted psychometric curve and SSQ data are also analyzed with ANOVA and the same post hoc method mentioned above.

3 RESULTS

In user subjective detection, we found significant main effects of *Spatial Size* ($F(1, 7526) = 16.741, p < .001$) and *Gain* ($F(5, 7526) = 7697.773, p < .001$). No significant difference was found in *Visual Density* ($F(1, 7526) = 0.380, p > 0.05$), *Realism* ($F(1, 7526) = 0.476, p > 0.05$), and *Gender* ($F(1, 7530) = 6.499, p > 0.01$). The post hoc pairwise t-test comparison revealed that the mean value of "greater" responses in the spatial size condition (*Landscape*) is higher compared to *Room*, especially in the gain values of 0.8 and 0.9, see Figure 2 Middle.

Subsequently, we estimated rotation gain thresholds in all scene conditions, and summarized the results of 25%LDT, 50%PSE, and 75%HDT in Figure 2 Right. To conclude, in 25% LDT and 50% PSE comparisons, we found significant main effects of *Spatial Size* with ($F(1, 210) = 10.026, p = 0.0017$) and ($F(1, 210) = 7.246, p = 0.007$) respectively. No significant difference was found in *Visual Density*, *Realism* and *Gender* in all 25% LDT, 50% PSE, and 70% HDT comparisons ($p < 0.05$). Post-hoc analysis showed the average 25% LDT and 50% HDT in four-room settings were higher (approaching 1.0) compared to four landscape settings. These results are concluded in Figure 2 Left. The graph's x-axis displays the rotation gains for the x-axis $g_{Rot}[x]$ in the VR scene. The graph's y-axis shows the probabilities of participants' responses to the 2AFC question. No interaction effects were found in either user subjective detection or estimated rotation gain thresholds at all.

Meanwhile, we followed the standard procedure [1] to calculate the SSQ total score to quantify the user's simulation sickness. We aggregated the SSQ scores over the four-day experiment and reported only the pre-experiments (pre), end-of-first-scene (middle), and end-of-second-scene (post) SSQ as two experimental blocks were carried out per day.

We measured a pre-SSQ score of 2.48 ($SD = 4.49$) and a middle-SSQ score of 15.83 ($SD = 15.02$) for the first VR scene, and 22.41

($SD = 23.20$) for the post-VR scene. There was a significant effect ($F(1, 48) = 18.149, p < 0.001$) for the pre-middle comparisons and ($F(1, 48) = 17.80, p < 0.001$) for the pre-post comparisons in the SSQ score. Post-hoc tests showed that the post-SSQ and middle-SSQ scores were significantly higher than the pre-SSQ score. ANOVA revealed no significance ($F(1, 48) = 1.418, p = 0.239$) in the score difference between the two blocks (middle-post comparisons, indicating that the users' states were equivalent at the end of both VR scenes. This result is consistent with the previous experiment [4].

4 CONCLUSION

Our work agrees with the conclusion from Paludan et al.'s [3] study, illustrating that one virtual environment (VE) can make another VE seem relatively 'slower.' In all users' subjective detections, the probability of 'greater' responses at gain values of 0.8 and 0.9 is higher in all room settings. ANOVA results show significant differences in the 25% LDT and 50% PSE when comparing the four landscape settings with their counterparts. The significant effects of visual density and realism factors are not found. No significant differences are observed in the 75% HDT in all tested conditions.

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